

EJOLT evaluation case study n°2

WP 4 and WP 8

Energy controversy over shale gas in Europe, with insights from North America

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Introduction

The aim of this case study is to describe the controversy related to the exploitation of non-conventional gas, or shale gas, based on the experience of the United States and the current state of the debate in Eastern and Western Europe. This controversy is viewed as a social choice dilemma cutting many issues such as: energy needs, health, environment and economic developed. We follow the general guidelines of the Integraal framework for sustainability assessment developed by REEDS research laboratory as a roadmap allowing us to deal with complex issues related to sustainability¹.

After describing the social choice dilemma behind the controversy, we identify the main issues at stake, the stakeholder groups that are concerned and the policy options proposed. A more thorough insight into the notion of social choice will lead us to broaden the notion of policy options to that of scenarios. We retain three scenarios reflecting broad visions of a desirable future, and their associated values. The last section gives an insight on the advantages of a deliberative approach to build collectively an issue to the controversy, as well as some general remarks on the way this approach could be deployed.

1. The Social Choice Problem

Unconventional sources of gas and oil seem to present for many countries tangible opportunities of a new, cheap and reliable source of energy, accessible through the modern technique of hydraulic fracturing (*fracking*), thus answering to rising demands for energy. Unconventional gas, mainly shale gas, now makes up about 60% of marketed production in the United States. Explorations have taken place in Poland, China and other countries. As shale gas is widely dispersed geographically, it also seems to offer economic development opportunities for local landowners and beneficiary communities. A new report for the US also highlights the potential of shale oil, still under investigation.

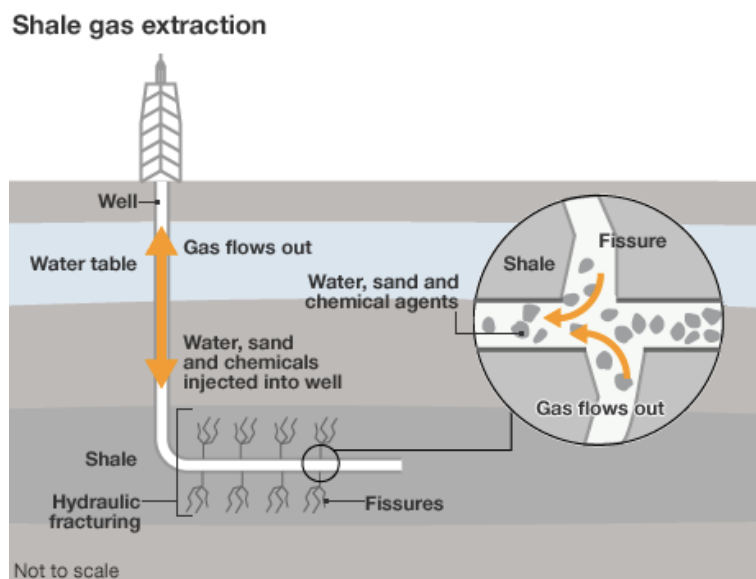
The International Energy Agency² issued a report in 2011, entitled “The Golden Age of Gas”, which underlines the economic potential of unconventional gas globally, as well as giving an analysis of the environmental impact, as compared to conventional gas, and indications of mitigation measures.

¹ For details of this framework, see: <http://tinyurl.com/Integraal-en>

² The IEA has 28 members, all from the geopolitical North, including Turkey as an emergent economy.

In the countries involved in exploration or exploitation, civil society groups and activists have underlined the risks involved in hydraulic fracturing (or fracking), the only technique available today for exploring and exploiting unconventional gas deposits, shale gas. This new technique “involves injecting huge amounts of water, mixed with sand and chemicals, at high pressures to break up rock formations and release the gas” (New-York Times, Feb. 26, 2011).

The diagram below gives a representation of the technique of hydraulic fracturing (fracking), which helps us understand the risks involved. We will give a brief summary or the important aspects of this technique.



Shale gas extraction involves drilling deep to the layer where the gas is trapped, and then horizontally in the shale layer (up to 3000 meters long). After drilling, water mixed with chemicals with different functions is injected into the drilling pipe, which will flow out under high pressure through holes in the pipe within the shale layer, in order to fracture the rock, liberating gas, which flows back in the pipe and out of the drilling well.

Despite allegations from industrials that the technique is safe and harmless, exploitation of this technique in North America has shown a variety of risks and damages to the local environment and communities. The main categories of risks or potential impacts identified are: water table and air pollution (through migration of gases and hydraulic fracturing chemicals), surface contamination from spills and flowback water, with the related impacts of those impacts on health, as well as additional greenhouse gas emissions. These categories of risks and impacts are discussed in the next section of this paper.

As a recent report underlines, some EU member States describe shale gas as “the most convenient bridge between a carbon-intensive past and a decarbonised future, while for others it is rather a source of security concerns.” (id.). This shows the depth of the gap between governments and actors, making a sustainability assessment necessary.

Policy review

A review of national and local policies regarding shale gas and hydraulic fracturing can prove useful to understand the level of public concern concerning the risks involved in shale gas exploration and exploitation as well as hydraulic fracturing.

Shale gas deposits are widely exploited in North America, with the USA at the forefront. The following figures offer a quick balance of the impact of this exploitation on the US territory: 700 000 drills have been done in the last 25 years, and 400 000 unconventional wells (80% of the gas wells in the country) are under exploitation. In 2005, the Energy Bill (or Energy Policy Act) allowed the oil and gas industry to bypass prior controls (for air, water quality, among others) by the Environmental Protection Agency before injecting chemicals in the ground.

Local communities, as well as some official bodies are assessing the consequences of a couple decades of exploitation, positive and negative. Lobbying groups, disseminating information through documentaries (see below), TV spots and other means, are advocating passionately usually for and for some against shale gas and fracking. Two states (New York and Maryland) have responded to sensitivity of their citizens to the risks of fracking for shale gas exploration and exploitation by a moratorium on fracking. Activity continues at a large scale in other states.

In Europe, the biggest reserves are in France and Poland. Each of those two countries have adapted opposing policies: while the French Parliament banned hydraulic fracturing in July 2011, Poland has given around one hundred exploration permits, including to large companies such as Chevron, Exxon Mobil and ConocoPhillips. Bulgaria is the second country to have banned fracking in January 2012, after demonstrations of opposition from civil society. After granting a few permits, the Czech Republic and Romania are facing opposition from civil society, and determining a position.

The following table summarizes the restrictions and ban on shale gas in a selection of major countries concerned by this issues related to shale gas and fracking. It gives an illustration of the level of public concern in the selected countries.

Table 1: Restrictions in shale gas exploration in selected countries

<i>Information</i>	<i>Type</i>	<i>Link</i>
United States (New York) In Dec. 2010 the governor placed a moratorium on high volume horizontal fracturing until July 2011	Moratorium	Source: WE02011 Golden Age of Gas Report
United States (Maryland) In March 2011 the house of representatives decided to place a moratorium on hydraulic fracturing until 2013.	Moratorium	Source: WE02011 Golden Age of Gas Report
Canada (Québec) In March 2011, the provincial government put shale gas exploration on hold until an environmental evaluation is conducted	Moratorium	Source: WE02011 Golden Age of Gas Report
France Temporary moratorium in June 2011 by Prime Minister. In July, the Parliament banned fracking. In October, the government revoked the 3 permits allowed for shale gas, for fear that the prohibited technique would be used. As of August 2012, the socialist government has expressed its commitment to the precautionary principle, and wants to reform the mine code (Code Minier).	Moratorium, then Ban	Source: WE02011 Golden Age of Gas Report Journal officiel du 14 juillet 2012 on http://legisfrance.gouv.fr

<i>Czech Republic</i>	The senate drafted a law in May 2012, banning shale gas extraction.	Draft law	http://www.ceskapozice.cz/en/news/politics-policy/czech-senate-drafts-law-banning-shale-gas-extraction
<i>Bulgaria</i>	In Jan. 2012, Parliament bans fracking for shale gas. In June 2012, the ban is partially lifted for conventional gas and oil exploration and extraction only.	Ban	http://www.bbc.co.uk/news/world-europe-16626580 http://www.guardian.co.uk/world/2012/feb/14/bulgaria-bans-shale-gas-exploration http://www.novinite.com/view_news.php?id=140297
<i>Romania</i>	Debates on banning or not shale gas		http://af.reuters.com/article/metalsNews/idAFL5E8G418Y20120504
<i>India</i>	In March 2011, differed by a year exploration of shale gas	Moratorium	Source: WE02011 Golden Age of Gas Report
<i>South Africa</i>	In April 2011, government placed an indefinite moratorium on hydraulic fracturing in the Karoo basin.	Moratorium	Source: WE02011 Golden Age of Gas Report

The next section will allow us to have an insight on the public concerns which may have motivated these restrictions on shale gas exploitation or exploration, by national or state authorities.

2. Issues, stakeholders, options and scenarios related to shale gas exploitation

The social choice issue and policy dilemma related to shale gas call for a social debate on the opportunities and drawbacks of exploiting this resource with the existing techniques, i.e. tapping into a newly identified source of energy that is widely dispersed and potentially profitable, both for private companies and local stakeholders, against facing different categories of risks and potential damages related to the use of the new and powerful technique necessary for the exploration and exploitation of unconventional gas.

To prepare this social debate, we have identified, following sustainability guidelines formalized by C3ED/REEDS, three necessary categories of analysis through a review of documents related to the controversy: the main issues at stake, the stakeholder groups involved, as well as the policy or scenario options expressing the terms of this controversy.

a. Issues at stake

The first category of analysis we analyzed is related to the performance issues, which give us a true insight into the costs and benefits of the exploitation of shale gas. Performance issues relate to all the dimensions of sustainability affected by the topic analyzed – exploitation of shale gas – from environmental to economic, social and political aspects.

Here, the concepts of “costs” and “benefits” are not necessarily measured on a monetary scale, but on a diversity of scales, to be defined through the assessment process, measuring impacts on well-being, health or the environment, as well as the economy.

The available information (see next section), allows us to identify some of the performance issues involved in the exploration and exploitation of unconventional gas:

- Energy availability and energy price

Under the current economic system, characterized by high metabolic activity³, energy demand in the next decade could surge by 35% from 2008 to 2035, according to the “Golden Age of Gas scenario” designed by WEO, 80% of that increase coming from non-OECD countries. As reserves of shale gas are fairly evenly distributed in the world (at least where exploration has been conducted), unconventional gas deposits seem to represent an opportunity for countries that will face high energy demand, without having significant reserves of conventional oil or gas. Reserves of unconventional gas are currently estimated to equal roughly those of conventional gas, amounting to 187 billion cubic meters, i.e. at least 60 years of world consumption.

³ "Contrary to the biological notion (...), this socio-ecological concept links materials and energy flows to social organisation, recognizing that the quantity of economic resource use, the material composition and the sources and sinks of the output flows are a function of socio-economic production and consumption systems that are highly variable across time and space." (from the web site of the Institute for Social Ecology, Alpen-Adria University, Vienna).

Local effect of unconventional gas exploitation is arguably positive on the energy bills of households, who are one of the primary consumers of gas, as well as industries seeking for cheap energy sources. In the United States, for example, prices of gas have dropped as a result of the exploitation of unconventional gas on a large scale.

- Economic opportunities for local landowners, for local inhabitants and business

Depending on national and local legislation, landowners can receive payments for the exploitation of shale gas under their land. In the USA, the Chesapeake Company is paying landowners 5 000 \$ an acre for the authorization to exploit shale gas.

The sustainability of job opportunities in the fracking industry needs to be assessed. While European environmentalist groups mention “boom and bust” cycles in local economies, studies in the USA highlight the creation of direct jobs and the development industries downstream of gas production.

In Texas, for example, shale gas is arguably creating further economic opportunities for the petrochemical industry, “which is converting cheap and abundant natural gas into resins and polymers for items like synthetic clothing and cellphones” (New York Times, July 16th, 2012). Sustainable economic impact has to be assessed on a case-to-case basis, along with the impacts of this downstream effect on other performance issues.

However, other economic impacts can be locally negative: low prices have led some smaller gas companies to bankruptcy, and facilitated mergers and acquisition by larger companies, the only ones able to pay for the contemporary fracking technique developed by transnational company Halliburton.

- Environmental quality and health risks

The main categories of risk identified, for the environment and/or for health are: greenhouse gas emissions, water quality, air quality, and local seismic activity.

Greenhouse gas emissions

- The gas extracted through fracking is methane, a gas 25 times more powerful as greenhouse gas (GGs) than carbon dioxide. Favourable studies argue that this is still less GGS than oil or coal combustion. Though companies try to limit leakage as much as they can, a certain level of gas leakage seems unavoidable. This phenomenon is responsible for more emission of methane than venting or flaring, but less than burning the gas, adding around 20% to the latter.

According to the WEO, shale gas emissions are 3 to 12% higher than the emissions of conventional gas. It is unclear if this evaluation takes into account the full range of emissions produced by the intense truck traffic necessary for production of the gas, through transportation of sand, water and chemicals, as well as removal of flowback water (see last criteria in this list).

Water

- The technique of hydraulic fracturing involves injecting large amounts of water (10 000 to 20 000 cubic meters per extraction), mixed with sand and chemicals, at high pressures.

More than two thousand five hundred types of chemicals have been used for fracking, according to the Chamber of Commerce of the US House of Representatives. 596 of those chemicals have been identified through the private initiative of scientist Theo Colborn, 29 of which are known to be damaging for health or the environment. Their effects on exposed population are one or several of the following: carcinogenic, mutagen, neurotoxic, toxic to reproduction, endocrine disruption. Many other chemicals used are kept as industrial secrets.

Local reports of health problems due to contaminated water are widespread in areas where shale gas is exploited in the United States. Contamination of water by gas also means there is fire hazard for local inhabitants of those areas.

The industry argues that the shale layers are separate from the water table, being hundreds of meters below the latter. But leaks of polluted and radioactive water, residues of the fracking process or of elements naturally present underground (like radium) have occurred. Risk on individual sites is multiplied by the number of sites under exploitation (hundreds of thousands). According to a study by the industry itself, around 6% of the cement pipes used in the exploration phase experience cracks, leading to a flowback of contaminated water into the pipes.

Air

- Air pollution can occur through several process: one is through the chemicals contained in the flowback water (known in the industry as “produced water”), amounting to roughly half the injected water, the other is through the gas refining process, which evaporates on site dangerous chemicals. This has been reported to produce locally high level of ozone concentration.

Seismic activity

- Other hazards. The fracking techniques also create local earthquakes. The effect of this seismic activity may be minor at the moment, however their impact also needs to be assessed, as they may damage gas wells for example. Stored condensates can contain explosive chemicals; explosions have been reported locally.

Local transportation

- The additional burden of large truck traffic (at least 20 000 cubic meters of water are transported at each injection) on road infrastructure may be a major factor of accident hazards (one accident a week involving a truck involved in shale gas is

accounted for in Pennsylvania) and higher maintenance costs for local authorities (it should be taken into account that companies have in certain states paid for upgrading roads, thus covering additional costs, but to what extent this is true and will last should be assessed).

- GHG emissions should also be appreciated in the full life cycle of shale gas. Its production involves the transportation of very large amounts of water by truck, which causes additional CO₂ emissions.

b. Stakeholder groups

Stakeholders are all people and organizations concerned by one or several aspects of shale gas exploration or exploitation. It is known that costs and benefits of a project or an activity are not evenly distributed to all stakeholders involved; what is an economic benefit for one stakeholder category can be a cost on health or wellbeing for another category. This is why the identification of stakeholders is necessary to a genuine and strong sustainability assessment. The method for assessing the impact using a multicriteria and multistakeholder method is considered at the deliberation stage (4). Our objective at the moment is to identify the main stakeholder groups.

Review of literature on the issues described above is an efficient method leading us to a list of stakeholder groups. We refer here to groups, or categories, of stakeholders, as individual people and institutions change from one location to another. The important, for the organisation of the evaluation process, is to identify which groups are concerned. Each group has a specific perspective and stand on the controversy; usually this is remarkably stable within a same country from one site to another.

Our review leads us to the following list of groups:

- **Local communities**, including land owners and non-land owners;
- **Local authorities**;
- **Government and policy makers**, including Energy departments, Environment and Water regulation bodies;
- **Energy companies**, including major companies involved in gas, local companies, companies involved in renewable energy;
- **Experts of different fields** (environment, economy, energy, social) ; and
- **Sources of information**: printed, radio and TV, online media, social groups, etc.

It is important to keep in mind this list of stakeholder groups, as each group has specific interests and perspectives on shale gas exploitation. Each of those is legitimate, as long as they can be argued through some type of rationality, and confronted to other points of view. Different stakeholders often refer to different types of rationality, also called “valuation languages” in the ecological economics literature. This makes the deliberation exercise particularly relevant for our evaluation purpose.

c. Policy options and scenarios

The controversy analysed in this case study relates to divergent and clearly-cut options regarding the exploitation or non-exploitation of shale gas. It is important for the assessment process to give a clear and complete description of each option. In other words, costs and benefits will be obviously different for each option, making it necessary to describe those thoroughly, especially in their contrasting features.

We will proceed in two steps in this section, first describing policy options, which are decisions aiming specifically at allowing or forbidding shale gas exploration and exploitation and its associated technique, and second scenarios, which refer to long-term options regarding energy and their implications on shale gas exploration and exploitation.

A first way to consider the official regulations regarding exploration and exploitation of shale gas is suggested by table 1 above. Those regulations refer to policies for or against the development of the shale gas industry. These policies are usually supported by statements or judgments referring to issues such as those identified above, and have the support of specific stakeholder groups, according to the issues that seem the most important to them. These positions can be discovered in the deliberation phase, when the assessment is done in a participative and deliberative way, or be induced by stakeholders' statements and publications.

Policy options

The main options available locally in operational terms for the short term seem to be one of two main options. A third option can be evoked, if we consider a larger time range, and the possible development of new techniques.

1. Exploiting unconventional gas deposits.

This option involves using the only existing technique available today: hydraulic fracturing, despite all the known issues and risks. This highly complex technique is described in several documents and documentaries in the list below.

2. Leaving unconventional gas deposits in the ground (in their geological layers).

This option is straightforward. At a larger scale, it is consistent with alternative energy policies, including the development of renewable sources of energy, as well as energy efficiency and sobriety measures.

3. A third option could be: **not exploiting shale gas until an alternative technique, proven clean and harmless, is developed** for exploiting them. This means prohibiting hydraulic fracturing, like France did in July 2011.

This option could have immediate implications, such as: inciting energy company to research new, harmless, techniques, for exploiting shale gas.

The three options considered above are what straightforward, or “no-brainer” alternatives. A deeper investigation into policies show a range of options in terms of accepting or not fracking at national, state or local level: moratorium, banning; prior public consultation could also be an option.

If we consider the full description of the controversy, as well as the issues at stake, we can imagine wider policy options, which we chose to call “scenarios”. These scenarios can allow us to link the controversy on shale gas and fracking with those on other energy sources, for example: exploiting oil from different areas (rainforest, tar sands, etc.). They are closely linked to the social dimension of the energy debate, involving societal choices at local and national level for a period of time longer than usually contemplated by the political agenda. Each of these scenario is underpinned by a choice of values, which is in its turn related to some fundamental parameters of social metabolism, such as per capita energy consumption, the evolution of which can only take place on the long term.

Scenarios

We have imagined the three following scenarios:

1/ The modernist and technical approach

This scenario is similar to the GAS scenario of the WEO, and involves exploiting all available gas deposits, trying to mitigate negative impacts for environmental and health. It considers the energy sector alone, setting energy availability as a top priority.

2/ The environmentally conservative approach

This scenario sets the preservation of environment as top priority. It involves not exploiting gas deposit that pose environmental and health problems. It implies assessing the consequences in terms of energy or economic development, but answers those issues as they come along at local or national level.

3/ The energy efficiency and sobriety approach

This scenario follows an integrated approach, not setting a single area of priority, but rather a set of priorities tending to well-being in harmony with nature. This set of priorities includes: protection of the environment, availability of energy and economic development. It takes into consideration the interactions between sectorial policies, as well as societal changes induced by policy as well as spontaneous evolution of values in society. For our matter, its main components are: downsizing production and consumption of fossil fuels, developing energy efficiency as well as energy sobriety.

While energy efficiency relates to the technical objective of minimizing the energy input for the production of a same good or service, energy sobriety is a multidimensional perspective that sets as shared objective the reduction of energy consumption and needs, adjusting it to a sustainable level acknowledging local and planetary boundaries.

3. Sources of information

Identification of relevant information is fundamental to organize any type of assessment exercise, be it through expert review or through participative dynamics, such as deliberation. Collection of information is the first step for developing indicators supporting the evaluation of the different options or scenarios with regards to the performance issues identified.

Information on shale gas exploration and exploitation and its impact is found in news briefs, documentaries, reports and official regulations. The selection below gives the main sources of information we have consulted for this paper.

a) Reports

The International Energy Agency: Are we entering a golden age of gas? Special Report, 20011, 131 p.

The introduction of this study gives a good summary of its perspective:

“The future for natural gas is bright. Demand has experienced a strong post-crisis recovery, while the North American shale gas boom and expansion of LNG trade have made ample supplies available in the near-term and bolstered future gas supply prospects. With mounting concerns over energy security and global climate change, and renewed debate surrounding the future role of nuclear power, these developments merit a deeper investigation of the prospects for, and the implications of, a golden age of natural gas.”

The report focuses mainly on market prospects (supply and demand) and technical aspects. Two sections deal with environmental impacts: Energy-related emissions (pages 37-41) and environmental impact of gas production and transport (pages 60-65). The first of these sections focuses on greenhouse gas emissions (CH₄ and CO₂ mainly). In the second section, the report underlines the public concerns regarding unconventional gas exploitation, related mainly to water consumption necessary for the fracking technique used for the exploitation of this type of gas, the contamination of the water tables, as well as the higher emission of CH₄. It puts emphasis on the existing techniques for mitigating those impacts, and produces the figures of 3.5 to 12% higher emissions in the worst case.

However, the real picture, in terms of social debate, is much larger than those elements. It is necessary to identify who are the stakeholders concerned, and document their concerns and perspectives on costs and benefits of shale gas exploitation.

b) Documentaries

Josh Fox, 2010: Gasland, documentary, 104". <http://www.gaslandthemovie.com/>

This documentary, based on interviews with people having allowed shale gas exploitation on their land, and field visits, reflects deep concerns with the exploitation of shale gas in the USA.

Shale gas and America's future, American Clean Skies Foundation, 30"
<http://www.shalegasfuture.com/>

This documentary presents itself as a balanced state of the debate on shale gas, but defends *in fine* this option as economically and environmentally sound. It is in fact an answer to the success of Gasland in the USA.

c) Dossiers

Affirming Gasland, July 2010, PDF, 39 p. is a dossier supporting arguments made in the film.

Chemicals in Natural Gas Operations
<http://www.endocrinedisruption.com/chemicals.introduction.php>

Marine Jobert François Veillerette: Le Vrai scandale des gaz de schiste, éditions Les liens qui libèrent, 2011.

d) Press releases and articles

European Unconventional Commercial Gas Potential Rivals North America, IHS CERA Study Says, Press Release, March 9th, 2011

"Regulation Lax as Gas Wells' Tainted Water Hits Rivers", New-York Times, February 26, 2011.

e) Position statements

Conclusions &Proposals – Workshop Marseilles – March 16th, 2012
Against fracking and shale gas, let's build a European international movement!
Food & Water Watch, ATTAC and others.

This is a declaration of different social movements and civil society organizations against the exploitation on shale gas.

Food & Water Europe, Friends of the Earth Europe, Greenpeace, Health & Environment Alliance, 2012: Position statement on shale gas, shale oil, coal bed methane and "fracking"

4. Insights for a deliberative approach of the shale gas controversy

4.1. The need for participation in assessing options and scenarios

This analysis on the need for participation in assessment of shale gas risks and opportunity will be procedural and not substantive. It aims at giving some guidelines for conducting consultation and deliberation activities on this question of social choice.

The following observation shows the need for a public debate:

“In 2010, European discussions about unconventional gas were limited to energy experts and the industry, but this year [2011] they were translated into political decisions, public campaigns and legal measures, such as ban on hydraulic fracturing in France” (Wyciskiewicz, 2011).

Despite public concern, decisions of public interest in Europe as well as elsewhere are often taken locally or nationally without public consultation, even for issues for which concern is high in society. In the case of shale gas, the vote of a moratorium or a ban by members of Parliament in France, South Africa, and some states of the US), appears more as a reaction to civil society actions or to media campaigns than as the result of the collective intelligence created by public debate. Proof is that part of the French government seem to want to reopen the possibility of shale gas exploitation, within the framework of the energy transition policy. The public and researchers will have more information after the environmental conference to be held on September 14th-15th.

Such a dynamic of action-reaction in public policies can bring concern about future developments, as it appears to induce case-by-case reactions to a controversy, which is not always followed by the necessary debate on energy policies. Civil society organizations believe that companies also have their own strategic plans at 3, 5 or 10 years, and may withdraw temporarily from a country, to sooth public concern, before coming back years later, when the context has evolved (Jobert and Veillerette, 2011). Thus, temporary moratoriums don't necessarily change their perspectives and mid-term plans.

Participation in sustainability assessment aims at having all stakeholders contribute to the debate on social choice. There are several ways of organizing a participative debate on an issue of collective interest such as shale gas exploitation. We have identified three main processes in the field of ecological economics: multi-criteria analysis, social multi-criteria evaluation, and deliberation. While all of these perspectives acknowledge the multiple dimensions of sustainability questions, what makes the difference between them is the process they suggest to explore this complexity and elaborate an answer that is satisfying from different perspectives. We have ordered them by decreasing degree of reliance on expert analysis and increasing level of participation.

- **Multicriteria analysis (MCA)** highlights multidimensional analysis of the social choice problem, “frequently using different measure scales, and a knowledge of the issue often suffering from a certain degree of uncertainty” (Vellejo, 2011).

While all processes of multicriteria assessment begin with a form of analysis, in pure MCA, the final evaluation is done through the use of formal tools, which can be defined as mathematical (such as: ranking, weighing, etc.), and is intended to lead to a univocal preference. The work of the coordinating expert, with his/her cognitive abilities to deal with multiple dimensions, is central to this process.

- **Social multicriteria evaluation** (SME) adds to multicriteria analysis a clear definition of stakeholder perspectives. Formalized by G. Munda in many publications, and disseminated in several countries, mainly Latin America, this perspective adds to multiple criteria multiple stakeholder perspective.

There are several ways of conducting a social multicriteria evaluation exercise, every one depending greatly on the facilitator's background, abilities, and preferred ways of operating. The taste of the exercise may vary from a collective cognitive exercise to a debate involving multiple dimensions.

- **Deliberation** is an open perspective on multicriteria assessment, which has participation of stakeholder to the evaluation process as primary rule. SME may lead to deliberation, provided that the exercise is participative enough.

In a typical deliberation exercise, as formalized by O'Connor, Douguet and others, all stakeholders are invited to express their preferences for the different options or scenarios, supporting their arguments regarding the different issues with units of information, or *indicators*, that are objective, measurable and comparable. Deliberation presents the advantage of providing a space for debate, which allows the expression of arguments for or against different policy options or scenarios, thus opening the opportunity for collective intelligence in questions of controversy and social choice.

The confrontation of arguments and the expression of values are meant to have multiple positive effects. The criteria for selecting options can be elaborated and agreed upon. New scenarios may emerge or existing ones be fine-tuned. Positions of stakeholders often evolve through this confrontation. Given the intricacy of the shale gas controversy with issues of economic development, environmental protection as well as energy transition, a deliberation perspective would arguably be much more efficient than a multicriteria assessment or a debate between experts.

4.2. Elements of method for deliberation

A methodological framework for conducting deliberation exercise in sustainability assessment has been formalized by the C3ED laboratory of Versailles-Saint-Quentin-en-Yvelines (now REEDS) research lab, in the first decade of the 21st century under the name "Integraal framework for sustainability assessment". We have followed the first steps of this framework for this case study. We give here a few methodological details that differentiate deliberation from other forms of multicriteria assessment. Those elements are expressed in part of step 3, and steps 4 to 6 of the Integraal framework. They include: 3/ Structuring information (after gathering it); 4/ Deliberation; 5/ Communication of the results and 6/ Reflection on the outcomes. For a presentation of each step of this framework, please refer to <http://tinyurl.com/Integraal-en>

Step 3 involves building a catalogue of indicators. A way to achieve this step is for the facilitation team to build (or help building) and suggest a catalogue of *candidate indicators* to the participants, which can be completed during the deliberation process (step 4) by other suggestions. Experience shows there is always in step 4 a need for the team to provide advice and often training on how to build quality indicators, or of a close interaction between the team and the participants during deliberation to elaborate indicators from the suggestions of the latter, which are formulated in a way that meets both scientific quality and acceptance by the participant with the initial concept.

During the deliberation exercise, each participant, identified as part of a stakeholder group, will evaluate costs and benefits of the different options or scenarios, according to each of the different performance issues. This evaluation can use one of the following options, similar for all participants: ranking alternatives, giving the preferred alternative, giving a discrete grade to each alternative, giving a continuous grade. According to the Integraal framework deliberation itself can follow one of the four types identified, through the combination of two choices between the following factors: either one or several participant per stakeholder group, and either the use of indicators to back-up position or no use of indicators.

4.3. Practical and axiological issues on MCE and deliberation

It is easier to have access to the theory of multicriteria evaluation (MCE) than to reports and lessons from MCE exercises effectively conducted on the field. Conducting deliberation in real life requires thorough preparation and facilitation. It is in our view a necessary step for improving deliberation methods. Following are some issues identified for shale gas exploitation through our experience as project manager and facilitator of international dialogues on development issues.

a) Uneven relevancy of performance issues for different stakeholder groups

A first observation completing the general guidelines of the Integraal framework for deliberation is the acknowledgement that all stakeholders may not be directly concerned by all the performance issues. This observation could affect the legitimacy of his/her vote on a scenario against this criterion. Stakeholders will scarcely agree on the fact that every criterion is equally important for them. But differences in valuation languages also mean that it is difficult to define which performance issues, or criteria, are most important. Participants from different stakeholder groups will believe that different sets of criteria are the most important for a particular evaluation.

Multicriteria evaluation frameworks are usually based on the acknowledgement of the non-substitutability of the different dimensions of sustainable development – known in economics as types of capitals, which are usually related to the different performance issues of deliberation. However, setting performance issues (“criteria”) as equally important is a question of social choice that goes beyond this postulate. It can be also argued that environment or public health be set as priorities through the mention of the precautionary principle, for example, as an argument emerging within the deliberation. This example shows that questions of evaluation are difficult to separate from questions of values. Economic development, well-being of local communities, the rights of nature, are perspectives usually backed up by separate sets of values, not easy to conciliate.

b) Differences in time and space scales and perspectives

The preferences of actors often refer to different time and space perspectives. For example, on a limited time scale, economic activity may be one of the positive results of shale gas exploitation, but analysis also shows a qualitative difference between bust and boom cycles and a sustained economic activity, each producing different effects on local and regional society and economy.

Environment impacts are often quite differentiated regarding space scale: greenhouse gas emissions are particularly preoccupying at global scale, being a shared responsibility for humankind, while water contamination often has a local, or at most regional impact on the environment as well as on human health.

Present in the environmental sphere, differences in time and space perspectives are particularly strong in the economic and social sphere. While scientific investigation is often available on environmental impacts and increasingly on health impacts, the economy remains a sphere of uncertainty, due to multiple ways and theories to apprehend it, and scientific apprehension of the social sphere is even more subject to controversy. Those differences in time and scale perspectives are related to differences in values. The precautionary principle is a clear illustration of this.

c) Shared values as a regulatory framework for deliberation

In order to arbitrate between votes it can prove necessary to refer to values recognized and shared by the whole society, i.e. across stakeholder boundaries. Reference to values or principles engraved in the constitution or fundamental law of a country could be a good proxy to those shared values. The precautionary principle, part of the Rio Earth Summit (1992), is for example acknowledged since 1995 in France, and enshrined in the 2005 Charter of the Environment, which is part of the Constitution. It is also acknowledged at the European Union level since the Maastrich Treaty (1992).

A social debate on shale gas in Europe, and particularly in France, could thus refer to this principle as expressing a shared value and a social consensus. Acknowledging the plurality of valuation languages and the diversity of stakeholder preferences is a necessary step, but this observation should not lead facilitators or stakeholders to forget that the social debate is taking place in a context where shared values have been acknowledged at a fundamental level. When those are enshrined in the constitution, they provide a convenient reference for arbitrating between preferences.

Conclusions

The controversy over the exploitation of shale gas in industrialized countries as well as in other contexts refers to a social choice problem involving a diversity of dimensions and issues. Because it cannot be solved by a linear and unequivocal approach, there is an opportunity to treat it through various methods of sustainability assessment. After describing the social choice problem related to shale gas exploitation, we found evidence of social and political concern in the restrictions on shale gas exploration and exploitation and in the risks and impacts of the associated technique of fracking.

Going through the first steps of sustainability assessment of shale gas exploitation led us to identify the main issues at stake in this controversy, namely: availability and price of energy, economic opportunities, quality of the environment and risks for health, impact of increased truck traffic. After identifying the main stakeholder groups concerned by one or several aspects of shale gas exploration or exploitation, we described three policy options for shale gas, and then three different scenarios, related to the future of energy.

After detailing the documentation supporting this analysis, we scrutinized different multicriteria assessment methods, with different levels of participation, ranging from expert analysis to deliberation. The latter seems more adequate to solve a question of social choice such as the shale gas controversy. We thus gave some details of the methods that can be used. Observing that deliberation is much more often theorized than applied, we offered some practical observations and conclusions on how to conduct a practical deliberation on the subject. We observed that, though languages of valuation are often different, reaching an agreement through deliberation implies that proper facilitation exists, and that the latter has in many cases the opportunity to refer to some shared values, which can often be found within the national legislation.

Hopefully this case study will help the researcher and the citizen to identify the stakeholders involved and the issues underlying the shale gas controversy. As the United States is now confronting a new energy dilemma with the discovery of huge deposits of oil in shale layers⁴, we hope it will help treat this controversy as a real social choice debate, which should be part of the energy policy vision of the countries involved. If shale oil can make the region “the new Saudi Arabia”, it seems particularly urgent to introduce a social debate on the exploitation of unconventional sources of fossil energy, and the associated techniques. Within the Europe Union, social debate is emerging in most countries where shale gas deposits have been found, evolving at different paces and with different outcomes in the member countries.

Sound methods for deliberation and debate over social choice are urgently needed to shed a light on those dilemmas, and suggest a mode of decision-making which takes into account the different categories of impacts, costs and benefits, risks, as well as the point of view of the stakeholders within this question of social choice.

⁴ Consultancies such as the Belfer Center for Science and International Affairs are predicting the “next revolution” of oil, with positive economic benefits. Exploration has started in Montana and North Dakota. After the US, other countries will face the same dilemma. US companies are already in contact with national companies to propose the exploitation of large shale gas deposits in China.

Links to the CEECEC glossary:

Commodity Frontiers, Corporate Accountability, Ecosystem Services, Peak Oil, Precautionary Principle, Sustainable Extraction, Uncertainty and Risk, Well-Being

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